

A Procedural MatCap System for Cel-Shaded Japanese Animation Production

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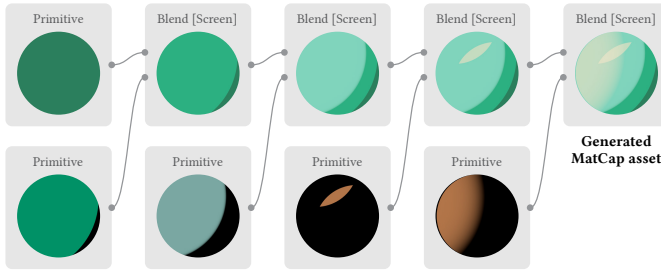


Figure 1: (Left) Concept of our *procedural MatCap* system. Unlike typical MatCap systems, it generates MatCap image assets in a fully procedural manner. (Right) Shaded models created by our procedural MatCap system.

ABSTRACT

MatCap is an expressive approach to shade 3D models and is promising for the production of typical Japanese-style cel-shaded animations. However, we experienced an asset management problem in our previous short film production; we needed to manually create many MatCap assets to achieve variations shot by shot or even frame by frame. In this work, we identify requirements for shading systems in Japanese animation production and describe our *procedural MatCap* system, which satisfies the requirements. Procedural MatCap generates customizable MatCap assets fully procedurally in run time, which drastically improves the asset manageability.

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1 INTRODUCTION

MatCap (material capture), also known as *Lit Sphere* [Sloan et al. 2001], is a useful and popular approach to shade 3D models in expressive and stylized ways. It does not simulate complex light

transport phenomena but directly picks shaded colors from a pre-defined raster image asset, which we call a *MatCap asset*, based on the camera-space surface normal. MatCap assets are often created by artists using painting software. One of the advantages of MatCap is the intuitiveness of its direct-color-picking mechanism, which allows artists to have a mental model of how to achieve the desired shading by manipulating the MatCap asset.

We as an animation production team believe that MatCap is a promising approach to the production of typical Japanese-style cel-shaded animations (hereafter denoted by Japanese animations) because of its intuitiveness (*i.e.*, easy to understand how to achieve the desired shading) and expressiveness (*i.e.*, capable of producing what artists want to express). In addition, the shading in Japanese animation production is often art-directed directly on the 2D camera viewport, and this is often prioritized over spatially consistent lighting [Todo et al. 2007]; in this sense, MatCap goes well with the design concept in Japanese animation.

However, the traditional MatCap approach is not practical in asset management when used in actual production. We took this approach to create a short film [2021], where we often needed to manually create new MatCap assets in a shot-by-shot manner, or sometimes even in a frame-by-frame manner, to achieve the best-quality animations. As a result, the number of assets we managed increased impractically, even though it was a “short” film.

Our Approach and Contribution. To overcome the limitation of MatCap while gaining its benefits, we designed and developed a *procedural MatCap* shading system (see Figure 1). We define procedural MatCap as the approach of generating MatCap assets in a fully procedural manner, without letting artists manually paint

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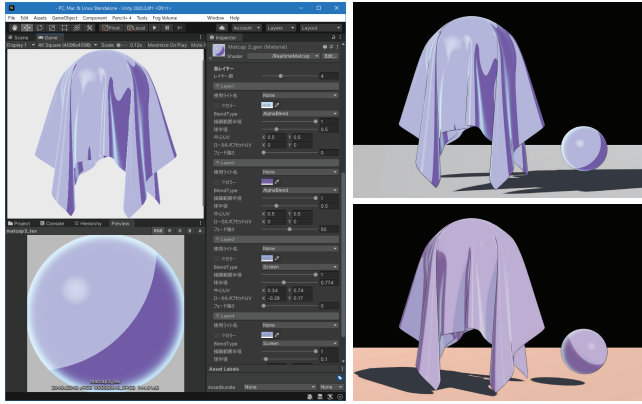


Figure 2: (Left) Our procedural MatCap asset design interface is implemented as a shader on Unity. (Right) Our MatCap assets can be parametrically controlled or even directly linked with light sources to achieve dynamic lighting.

MatCap assets. We describe the design considerations of our system and how this approach is suited to Japanese animation production.

Note that procedural MatCap itself may not be novel (e.g., *Todo et al. [2013]* demonstrated procedural manipulation on MatCap assets). Our contribution is in the discussion of the practical use of procedural MatCap in Japanese animation production.

2 SYSTEM REQUIREMENTS AND OBJECTIVE

Based on our experience, we identify three system requirements:

- R1: Directness** The system should allow artists to design the shading directly without being unnecessarily constrained by spatially consistent lighting.
- R2: Expressiveness** The system should be capable of producing various expressions used in typical Japanese animations.
- R3: Asset manageability** The system should avoid requiring artists to manually create an intractable number of MatCap assets when the artists make variations.

The use of MatCap satisfies **R1** and **R2**. We emphasize that **R3** is critical to production, where many shots need to be created. Besides satisfying these requirements, we consider **O: Simplicity** as our objective, meaning that we want to keep the system as simple as possible. Animation production involves many artists from different backgrounds, and so the learning cost needs to be minimized.

3 OUR PROCEDURAL MATCAP SYSTEM

Taking **R1–R3** and **O** into account, we designed our procedural MatCap system. The basic idea is to combine several *primitives* with adjustable parameters to generate a MatCap asset. The values of these parameters can be set shot by shot or even keyframed. MatCap assets are dynamically generated in run time and stored only on memory. This drastically reduces the necessity to duplicate MatCap assets for creating variations, by which **R3** is satisfied.

Considering the objective (**O**), we intentionally restrict the choice of primitives to only a circle shape. That is, all MatCap assets in our system are designed by combining circles in different ways. Each circle primitive is adjusted by the following parameters: color,

Table 1: Comparison to alternative shading approaches.

Requirement (see the text for details)	R1: Direct	R2: Express	R3: Manage
Ramp shader	No	Limited	Yes
Traditional MatCap [Sloan et al. 2001]	Yes	Yes	No
Extended MatCap [Todo et al. 2013]	Yes	Yes	Limited
Our procedural MatCap	Yes	Yes	Yes

blend type (e.g., normal, multiply, and screen), circular canvas radius, circle radius, center position, offset from the center of the canvas, and fade. We also restrict the way of combining circles to sequential layer overlay rather than allowing arbitrary structures. We empirically found that, though restrictive, this system design can offer sufficient expressiveness (**R2**) given that the target is Japanese animations. Note that artists can use painted MatCap assets directly as a fallback when creating irregular effects.

We implemented the procedural MatCap system on Unity. It consists of two shaders: a shader for generating a MatCap asset and writing it into a render texture (see Figure 2 (Left)), and a shader for applying the rendered MatCap asset to scene objects. Note that we add contours in postprocess separately from shading.

Dynamic Lighting. Traditional MatCap cannot be affected by scene lights. To overcome this limitation, *Todo et al. [2013]* proposed a method of dynamically manipulating MatCap assets based on the scene light direction. Since our MatCap is fully procedural, we can easily integrate this feature as an option (see Figure 2 (Right)). When this feature is enabled for a circle primitive, its center position is dynamically set from the scene light direction. So far, this feature has been frequently used by artists in scenes with dynamic lighting.

Comparison. Table 1 compares our procedural MatCap and other alternatives. Ramp shader is a basic approach to achieve cel-shading effects, but its expressiveness is not satisfactory. Extended MatCap [Todo et al. 2013] partially supports procedural operations, which improves asset manageability, but it still requires artists to create static MatCap assets manually. Ours satisfies all requirements.

4 CONCLUSION

We described requirements for shading systems in Japanese animation production and how our procedural MatCap system satisfies them. The system has already been used in the upcoming animation production, and it has been appreciated by experienced artists.

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